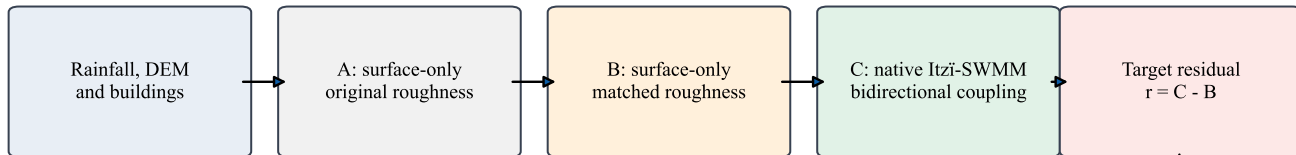
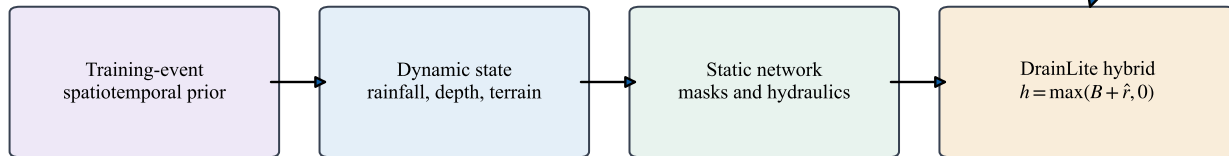


Paired physical controls and leakage-controlled residual learning



Physical decomposition: B-A isolates the inlet-neighbourhood roughness change; C-B isolates sewer coupling.



MIKE fields are used only as an external descriptive reference. The public LarNO checkpoint is reproduced separately.